FACULTY OF COMPUTING
UTM Johor Bahru

SECJ 2203: Software Engineering

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PROJECT PROPOSAL

XYZ SYSTEM

ABC ORGANIZATION

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1. Introduction

Firstly, what is a Quality Management System (QMS)? It is actually a formalized system that documents the processes, procedures, and responsibilities for achieving an organization's quality policies and objectives. The QMS main purpose is to ensure that one organization can consistently meet all the customer and regulatory requirements and continue to pursue future improvement. For companies like Company XYZ, a QMS that complies with the ISO 9001:2015 standard is critical for operations excellence and maintaining international certification.

Need : What system XYZ needs is a dedicated QMS software because their current system, which is a combination of shared documents, spreadsheets and many more types of files are inefficient and impractical. Specifically, there is a huge demand for a centralized system to manage internal audits, control documents, track customer complaints, and ensure timely equipment calibrations. Currently, these things are being tracked normally, and they are very time-consuming which sometimes leads to it being difficult to trace. Moreover, it also makes it difficult to prove compliance during external audits and increasing the risk of losing their ISO certification.

Approach : The proposed solution, "QMS-flow " is a web-based application that is designed with a modular architecture to address specific ISO 9001:2015 requirements. The system will be built on a modern technology stack with a centralized database, which will provide role-based access to employees, managers, and quality personnel. Key models will include :

1. Audit Management : To schedule audits, log findings, and assign corrective actions.
2. Change Control : To manage document versions that will avoid multiple duplications and enforce approval workflows.
3. Customer Complaints & CAPA : To log complaints, link them to corrective actions, and guide users through the root cause analysis.
4. Employee Training & Equipment Management : To maintain training matrices and track calibrations schedules with automated alerts.

Benefits : The implementation of a properly functional QMS-flow will transform the company's quality processes. It will also provide a single source of truth for all QMS activities, drastically reducing manual effort, which will also reduce human error. The system

will automatically send reminders if there are any overdue actions, upcoming calibrations, enhancing compliance. Ultimately, by digitizing and automating important workflows, QMS-Flow will save a lot of time, strengthen the organization's quality culture and secure its ISO 9001:2015 certification.

Competitors : Currently, Company XYZ does not use any dedicated QMS software. They highly rely on a disjointed set of generic tools like shared network drives, Microsoft Excel and many more types of different workspace to manage their quality processes. While other generic platforms like SharePoint can handle document storage, they lack the specific workflows, reminders, and integrated modules required for ISO 9001:2015 compliance. There are other QMS software products on the market, but they are often expensive and not tailored for medium-sized manufacturers. Our proposed QMS-FLow system is designed to be a cost-effective, integrated, and specialized solution that directly addresses the unique gaps in their current process, which moving them from a reactive, manual system to a proactive, automated one.

2. Existing Systems

As described in the case study, Company XYZ Sdn. Bhd. is a predominantly non-computerized, manual system. It relies on a fragmented combination of shared network drives for storing digital files, spreadsheets (e.g., Microsoft Excel) for tracking audits and complaints, and physical paper trails for documents like signed approval forms and calibration certificates. There is no central system connecting these different pieces, which creates significant gaps in their quality management process.

In the broader market, several commercial Quality Management System (QMS) software solutions exist. For this analysis, we have examined two established competitors which are Sparta Systems, MasterControl and Qualio. These systems represent the type of solutions available but highlight the gaps that QMS-Flow aims to fill for a medium-sized enterprise like Company XYZ.

To understand the competitive landscape, we have analyzed three established QMS platforms:

1. **Sparta Systems TrackWise:** A highly configurable enterprise-level QMS known for its use in heavily regulated industries like pharmaceuticals.

2. **MasterControl:** A popular platform focused on document control and process automation, widely used in manufacturing and life sciences.

3. **Qualio:** A modern, cloud-based QMS that emphasizes user experience and is geared towards small to mid-sized businesses.

The following table provides a direct comparison of their key features.

Features	Sparta Systems	MasterControl	Qualio
Centralized Document Control	Yes	Yes	Yes
Automated Audit Management	Yes	Yes	Yes
Integrated CAPA from Complaints/Audits	Yes	Yes (Modules)	Limited
Built-in Calibration Management	Yes (Add-on)	Yes (Add-on)	No
Drag-and-Drop Custom Form Builder	No	No	No

Training Matrix & Record Management	Yes	Yes	Basic
Real-time Dashboard for Management Review	Yes	Yes	Limited
Primary Target Market	Large Enterprise	Mid to Large	Small to Mid

Despite the availability of these software solutions, our analysis reveals two critical shortcomings that justify the development of "QMS-Flow."

Firstly, enterprise-grade systems like Sparta Systems are often prohibitively complex and expensive for mid-market companies. Their implementation can be lengthy and require specialized consultants, placing them out of reach for organizations like Company XYZ that need a more straightforward and cost-effective solution. The high complexity can also lead to poor user adoption among employees who are not full-time QMS specialists.

The second issue is a lack of tailored functionality for mid-market needs. Modern cloud-based systems like Qualio excel in usability but may lack the depth required for a manufacturing environment. The absence of a built-in calibration module is a significant drawback, forcing companies to manage a critical compliance area separately. Moreover, none of the analyzed competitors offer a simple drag-and-drop form builder, a feature that would empower Company XYZ to adapt their CAPA and other processes without needing IT support or expensive customizations.

Therefore, a clear gap exists for an integrated QMS that is both comprehensive enough to handle all core ISO 9001:2015 modules out-of-the-box and designed with the usability and budget of a medium-sized business in mind. "QMS-Flow" is positioned to fill this gap by

offering a unified, user-friendly platform that eliminates the need for costly add-ons and complex configurations.

3. Proposed System

3.1 Overview

The proposed system, QMS-Flow, is a centralized web-based Quality Management System (QMS) designed for Company XYZ to ensure compliance with the ISO 9001:2015 standard. It replaces the manual, spreadsheet-based process with a digital platform that integrates all quality management activities into a single, accessible environment.

3.2 System Objectives

1. Digitize quality management workflows (audits, CAPA, training, complaints).
2. Improve traceability and compliance through centralized record keeping.
3. Enhance productivity with automation and notification functions.
4. Strengthen accountability by assigning actions and tracking completion.

3.3 Key System Modules

The proposed QMS-Flow system is built using a modular architecture that aligns with the ISO 9001:2015 requirements. Each module performs a specific quality-management function but connects seamlessly to form a single, integrated platform. Together, these modules automate manual processes, improve traceability, and enhance compliance across the organization.

Audit Management enables the planning, execution, and follow-up of both internal and external audits. Quality managers can create annual audit schedules, assign auditors, and use digital checklists based on ISO clauses. Audit findings are logged directly in the system, and corrective actions are automatically assigned to responsible users with email reminders until closure.

Change Control provides a secure document repository where all controlled documents—such as Standard Operating Procedures (SOPs) and policies—are stored with

full version control. When a document change is requested, the file is locked for editing and routed through an automated review-and-approval workflow. Every action is captured in an audit trail, ensuring accountability and document integrity.

Customer Complaints offers a structured process to record and track customer issues. Any employee can log a complaint, which is then assigned to the quality manager for investigation. Each complaint can be linked to a Corrective Action record, enabling the organization to identify root causes and verify that the issue has been effectively resolved.

Corrective and Preventive Action (CAPA) is the central module for managing non-conformities. It supports root-cause analysis tools such as the 5 Whys and Fishbone Diagram, allowing users to investigate issues thoroughly. Each CAPA record includes target dates, responsible personnel, and effectiveness verification tasks to ensure long-term improvement.

Management Review automates the scheduling and documentation of management-review meetings. Agendas, discussion points, and decisions are recorded within the system, and follow-up actions are assigned to specific users with due-date reminders.

Employee Training maintains a complete record of employee competencies and training activities. It allows administrators to assign mandatory training by role or department, record completion details, and send reminders when refresher training is due.

Equipment and Calibration Management together ensure that all instruments and devices are properly maintained and calibrated. The system stores equipment information, schedules calibration activities, uploads calibration certificates, and issues alerts before calibration deadlines. These features help the organization avoid compliance risks related to outdated or unverified equipment.

In combination, these modules make QMS-Flow a comprehensive and scalable digital platform that transforms quality management from a manual, document-heavy process into a streamlined, automated, and transparent workflow.

3.4 User Roles and Permissions

- Admin: Manage system settings and user accounts.
- Quality Manager: Oversee audits, CAPAs, and management reviews.
- Department Head: Approve document changes and manage assigned findings.
- Employee: Access SOPs, record training, and log complaints.

3.5 System Advantages

- Real-time monitoring of QMS activities
- Reduced human error and paperwork
- Improved compliance with ISO standards
- Better decision-making through dashboards and analytics

4. Software Process Model

4.1 Justification on Choice

The Agile Scrum software process model is chosen as the most suitable approach for developing QMS-Flow, a centralized web-based Quality Management System designed to manage ISO 9001:2015 processes such as audits, CAPA, document control, training, and complaint management.

Agile emphasizes collaboration, flexibility, and continuous improvement, which align perfectly with QMS-Flow's iterative design and compliance-driven requirements. This model allows the team to adapt quickly to evolving user and compliance needs while maintaining high-quality standards.

Agile promotes incremental delivery, enabling the team to release functional modules in shorter cycles, gather feedback from users, and refine continuously. Regular stakeholder involvement ensures that every increment meets organizational goals and ISO 9001 compliance expectations.

Reason	Explanation
Flexibility	Requirements may evolve as Quality Managers and Auditors test early prototypes. Agile allows quick adaptation after each sprint.
Incremental Delivery	Each ISO-related module (Audit, CAPA, Document Control, etc.) can be developed and delivered in separate sprints, ensuring visible progress and early validation.
Risk Reduction	Frequent testing and stakeholder reviews minimize the chance of developing non-compliant or redundant features.
Quality Focus	Agile's sprint review and retrospective cycles align with continuous improvement principles, ensuring the system improves over time.
Team Fit	The four-member team fits the Scrum framework, with distinct roles that encourage collaboration and accountability.

Agile (Scrum) provides adaptability, transparency, and effective feedback cycles, making it ideal for QMS-Flow's modular structure and compliance-oriented objectives.

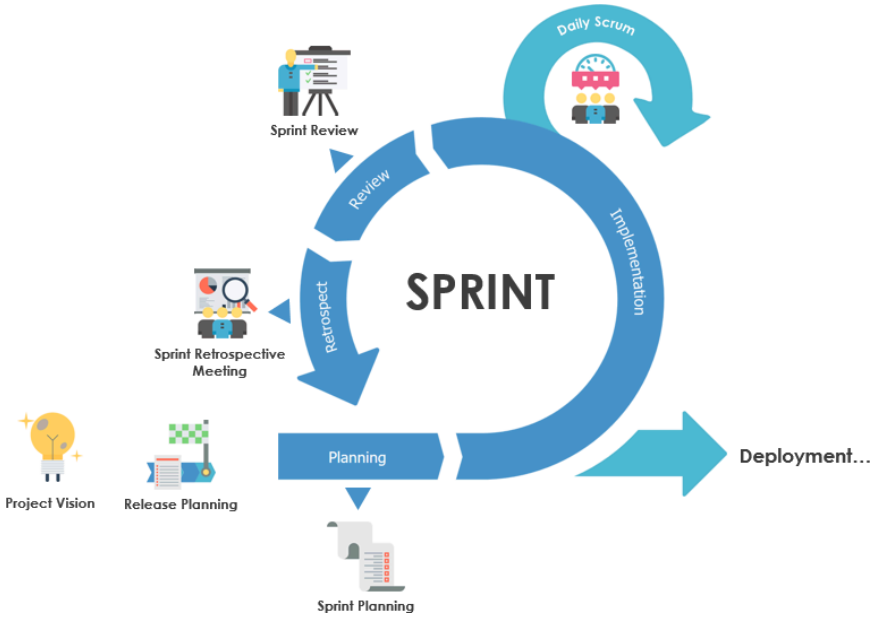
4.2 Detailed Activities

Development of QMS-Flow will be carried out in six Agile Sprints, each lasting 2 to 4 weeks. Every sprint includes planning, development, testing, and review activities to ensure consistent progress and improvement.

Team roles are assigned as follows:

Role	Team Member	Responsibility
Scrum Master	Seah Zhang Jian	Facilitates meetings, tracks sprint progress, removes blockers, and ensures smooth coordination.
Product Owner	Muhammad Luqman	Defines system requirements, manages the product backlog, and prioritizes user stories according to

		stakeholder needs.
Developer 1	Lee Hui Zhen	Designs, codes, and integrates system modules, assists in testing and documentation.
Developer 2	Tan Hui Shan	Develops and implements system features, conducts testing, and supports documentation tasks.



Sprint Breakdown

Sprint	**Focus Module**	**Activities**	**Responsible Roles**
Sprint 1	Audit Management	Define user stories, design UI mockups, and implement audit scheduling and findings logging.	Product Owner + Developer 1
Sprint 2	Change Control	Build document repository, version control, and approval workflow.	Developer 2 + Scrum Master
Sprint 3	CAPA & Complaints	Implement CAPA initiation, Root Cause Analysis (RCA) tools, and complaint linkage.	Developer 1 + QA Tester

Sprint 4	Training & Equipment	Develop training matrix, evaluation tracking, and equipment registry.	Developer 2 + Product Owner
Sprint 5	Calibration & Management Review	Add calibration alerts, meeting scheduler, and management review tracking features.	Scrum Master + Developer 1
Sprint 6	Integration & Testing	Integrate all modules, perform system testing, and prepare deployment plan.	All Members

Each sprint follows the same Agile cycle:

- Sprint Planning: Define backlog items and assign team responsibilities.
- Daily Standups: Short meetings to track progress and resolve blockers.
- Sprint Review: Demonstrate completed features to stakeholders for feedback.
- Sprint Retrospective: Reflect on performance and identify areas for improvement.

4.3 Deliverables and Duration

Sprint	**Duration**	**Key Deliverables**
Sprint 1 – Audit Management	3 weeks	Audit module prototype, backlog updates, user stories.
Sprint 2 – Change Control	3 weeks	Change Control module with document version control and approval workflow.
Sprint 3 – CAPA & Complaints	3 weeks	CAPA and Complaint Handling modules integrated into the system prototype.
Sprint 4 – Training & Equipment	3 weeks	Training management and equipment tracking modules.
Sprint 5 – Calibration &	2 weeks	Calibration module, meeting scheduler, and management review tools.

Management Review		
Sprint 6 – Integration & Testing	2 weeks	Fully integrated system demo, testing report, and deployment plan.

Total Duration: 16 weeks (including 2-week buffer for final documentation and review).

Final Deliverables:

- **Product Backlog:** A prioritized list of user stories and epics, refined throughout the sprints to reflect evolving requirements.
- **Sprint Reports:** Documentation of sprint goals, completed tasks, challenges faced, and retrospective insights.
- **Functional Modules:** Working software components for each QMS domain (Audit, CAPA, Training, etc.), validated through stakeholder reviews.
- **Integrated System Demo:** A cohesive prototype demonstrating interoperability across modules, suitable for final presentation and evaluation.
- **User Manuals:** Instructional guides for end-users and administrators, supporting training and system adoption.
- **Maintenance Plan:** A roadmap for future enhancements, bug tracking, and long-term system sustainability.

5. Project Schedule

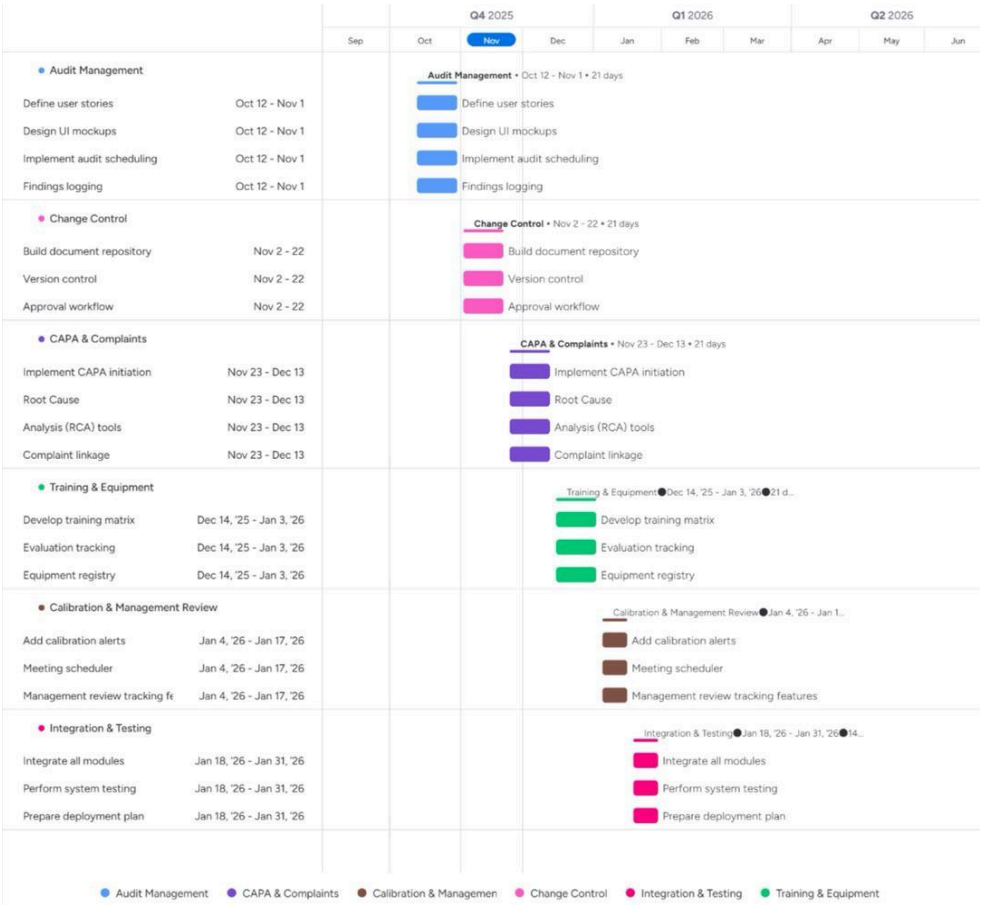


Figure 1: Gantt Chart

[URL : https://graduate62501.monday.com/boards/5024068175/views/49140359]

6. References

[1] Sommerville, I. (2016). “Software Engineering”, 10th Edition, US: Pearson.

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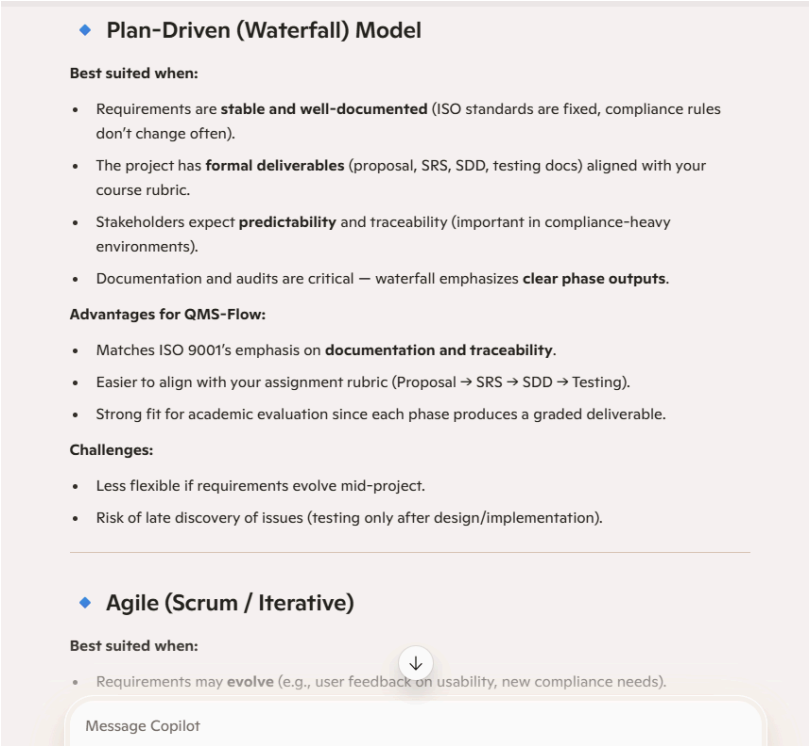
Appendices

Table 1: Comparison of Existing Systems and Proposed QMS-Flow

Features	Sparta Systems TrackWise [1]	MasterControl [2]	Qualio [3]	Proposed QMS-Flow
Web-Based Platform	Yes	Yes	Yes	Yes
Audit Management with Automated Tracking	Yes	Yes	Yes	Yes
Document Control with Version Locking	Yes	Yes	Yes	Yes
Automated Complaint-to-CAPA Linking	Manual	Manual	Manual	**Yes**
Custom CAPA Form Builder (Drag-and-Drop)	Complex	Limited	No	**Yes**
Integrated Calibration Alert System	Yes	Yes	No	**Yes**
Focus & Typical Customer	Pharma/Life Sciences	Medical Device	Startups	**M'sian Manufacturing**
Implementation Complexity & Cost	High	High	Moderate	**Low**

Figure: Comparison of Existing Systems and Proposed QMS-Flow



Plan-Driven (Waterfall) Model

Best suited when:

- Requirements are **stable and well-documented** (ISO standards are fixed, compliance rules don't change often).
- The project has **formal deliverables** (proposal, SRS, SDD, testing docs) aligned with your course rubric.
- Stakeholders expect **predictability** and traceability (important in compliance-heavy environments).
- Documentation and audits are critical — waterfall emphasizes **clear phase outputs**.

Advantages for QMS-Flow:

- Matches ISO 9001's emphasis on **documentation and traceability**.
- Easier to align with your assignment rubric (Proposal → SRS → SDD → Testing).
- Strong fit for academic evaluation since each phase produces a graded deliverable.

Challenges:

- Less flexible if requirements evolve mid-project.
- Risk of late discovery of issues (testing only after design/implementation).

Agile (Scrum / Iterative)

Best suited when:

- Requirements may **evolve** (e.g., user feedback on usability, new compliance needs).

Message Copilot

Figure: Choosing Between Agile and Plan-Driven Models

URL

1. https://chat.deepseek.com/a/chat/s/229899e3-9e77-415e-bf49-e7889ad2f15f
2. https://www.researchgate.net/publication/272241155_The_effectiveness_of_the_ISO_9001_quality_management_system_and_its_influential_critical_factors_in_Greek_manufacturing_companies
3. https://copilot.microsoft.com/shares/qzC3UbEiUWsTtt1FGCsgX

Marking Criteria	*Marks*
Introduction • *Needs* • *Approach* • *Benefits* • *Competitors*	*12 marks*
Existing Systems	*4 marks*
Issues or problem with existing systems	*4 marks*
Proposed System	*10 marks*
Software Process Model • *Justification on choice*	*5 marks*
Software Process Model • *Detailed Activities*	*10 marks*
Software Process Model • *Deliverables and Duration*	*5 marks*
Project Schedule	*5 marks*
References	*2 marks*
Overall report quality	*3 marks*
Report Total marks	*60 marks/5%*
Team-working (Peer evaluation)	*0.5 %*
Team-working (Instructor observation)	*0.5%*